

Super-Eddington Little Blue Dots Over-Reionize Helium

JWST's faint AGN may complete helium reionization too early unless super-Eddington accretion is rare.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint broad-line AGN at $z \gtrsim 4$, dubbed Little Blue Dots. Some models link them to Little Red Dots via super-Eddington accretion and viewing angle.

Whether super-Eddington LBDs can reionize helium without violating the observed end at $z \approx 3$.

What this paper is selling

The paper tests helium reionization timing under super-Eddington LBD scenarios. It finds that only $\lesssim 10\%$ of LBDs can be super-Eddington, assuming they dominate the helium-ionizing budget.

Key results to remember

- Super-Eddington LBDs would finish helium reionization too early, violating $z \approx 3$ constraints.
- Only $\lesssim 10\%$ of LBDs can be super-Eddington if they dominate helium reionization.
- Escape fractions $\lesssim 1.5\%$ or lower abundance would be needed for mostly super-Eddington LBDs.

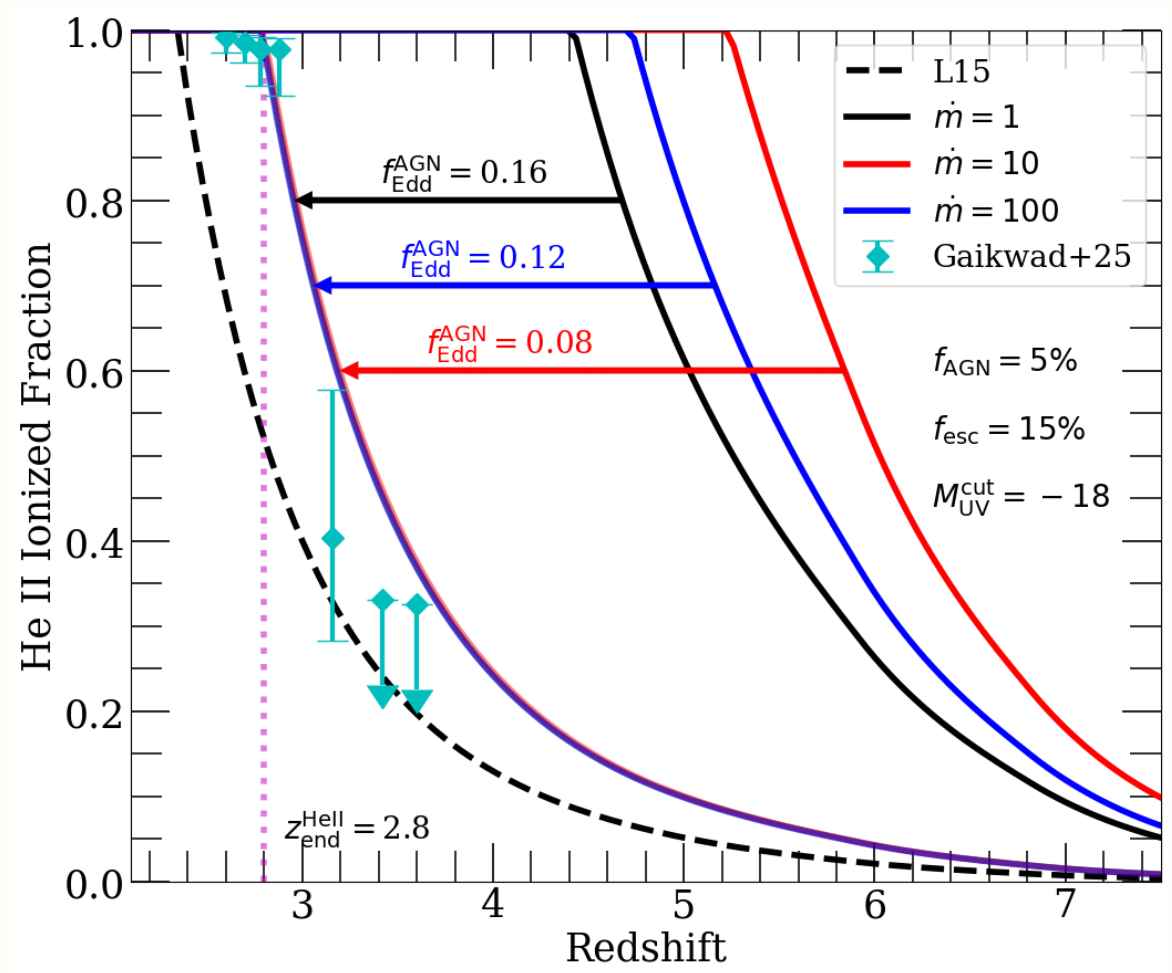


Fig. 3. Helium reionization histories for different SEDs; dashed L15 and solid soft-state models...

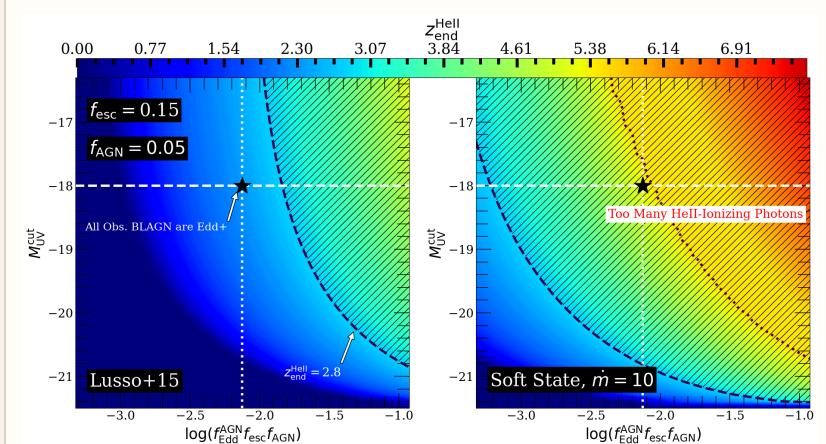


Fig. 4. Excluded AGN parameter space from helium reionization constraints; hatched region shows disfavored...